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# Methods Practice SAC1 - Part A

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No notes, No CAS

5m Reading | 50m Writing

32 marks

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## Question 1 (3 marks)

VCAA 2018 Exam 1 Q1

- a. If  $y = (-3x^3 + x^2 - 64)^3$ , find  $\frac{dy}{dx}$ .

1 mark

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- b. Let  $f(x) = \frac{e^x}{\cos(x)}$ .  
Evaluate  $f'(\pi)$ .

2 marks

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**Question 2 (3 marks)**

State the transformation that takes the graph of:

$$y = 2\sqrt{2-x} + 3 \quad \text{to} \quad y = -5\sqrt{x} + 6$$

Give your answer in the form:

$$(x, y) \rightarrow (ax + b, cy + d)$$

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**Question 3 (5 marks)**

VCAA 2018 Exam 1 Q7

Let  $P$  be a point on the straight line  $y = 2x - 4$  such that the length of  $OP$ , the line segment from the origin  $O$  to  $P$ , is a minimum.

- a. Find the coordinates of  $P$ .

3 marks

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- b. Find the distance  $OP$ . Express your answer in the form  $\frac{a\sqrt{b}}{b}$ , where  $a$  and  $b$  are positive integers.

2 marks

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**Question 4 (7 marks)**

Neap 2020 Exam 1 Q5

a. Solve  $1 - 2\sin(2x) = 0$  for  $x \in \left(-\frac{\pi}{2}, \pi\right]$ .

2 marks

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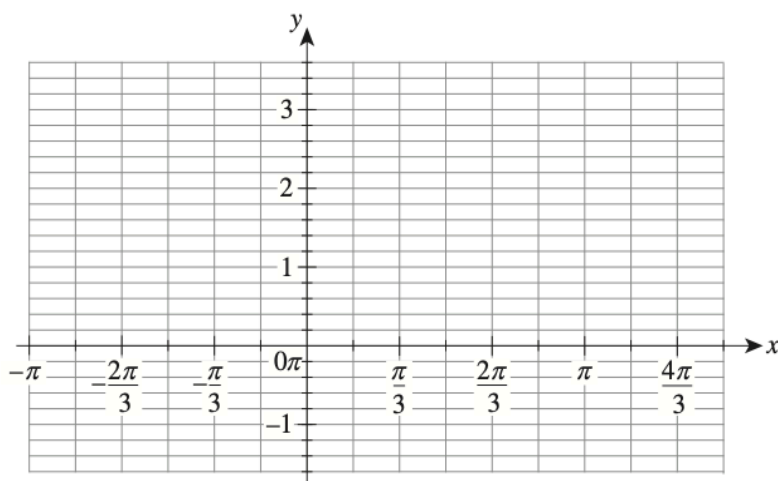
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Let  $f: \left(-\frac{\pi}{2}, \pi\right] \rightarrow \mathbb{R}, f(x) = 1 - 2\sin(2x)$ .

- b. Sketch the graph of  $y = f(x)$  on the axes below. Label the axes intercepts and endpoints with their coordinates as appropriate.

3 marks



- c. Find the equation of the tangent to the graph of  $y = f(x)$  at the point where the graph crosses the y-axis.

2 marks

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**Question 5 (5 marks)**

Solve the following equations for all **real** solutions,  $x \in R$ .

a.  $3^x + 2 = 15 \times 3^{-x}$

(2 marks)

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b.  $2\log_{10}(x) - \log_{10}(x + 6) = 0$

(3 marks)

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**Question 6 (4 marks)**

Consider the piecewise function,  $f$ :

$$f(x) = \begin{cases} x^3 - 3 \cdot x + 9, & x \leq 0 \\ a(x+a), & x > 0 \end{cases}$$

For what value(s) of  $a$  is the function  $f$ ...

a. **Continuous** for all  $x \in \mathbb{R}$  ?

(2 marks)

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b. **Differentiable** for all  $x \in \mathbb{R}$  ?

(2 marks)

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**Question 7 (5 marks)***Neap 2020 Exam 1 Q7*

Let  $f: \mathbb{R} \rightarrow \mathbb{R}, f(x) = e^{3x} - 2$ .

- a. Find the rule and domain of the inverse function  $f^{-1}$ .

2 marks

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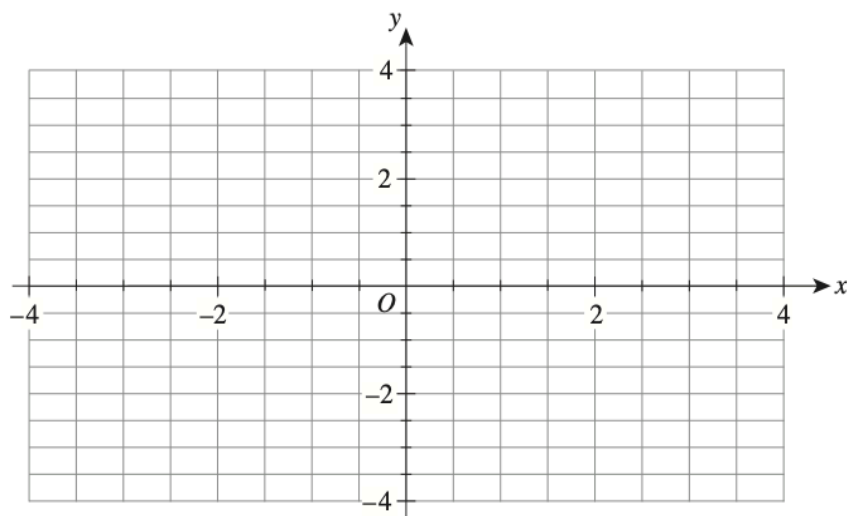
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- b. On the axes below, sketch the graph of  $y = f(f^{-1}(x))$  for its maximal domain.

1 mark



- c. Find  $f(f^{-1}(3x))$  in the form  $\frac{ax+b}{cx+d}$ , where  $a, b, c, d \in \mathbb{Z}$ .

2 marks

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